

CM0832 - MT5001 Elements of Set Theory

Introduction

Andrés Sicard-Ramírez

Universidad EAFIT

Semester 2026-1

Outline

Pácto Pedagógico

Introduction

First-Order Theories

References

Outline

Pácto Pedagógico

Introduction

First-Order Theories

References

Pacto Pedagógico

Como miembros de la Universidad EAFIT, nos comprometemos a actuar de manera íntegra siguiendo los más altos estándares éticos y morales.

- Respeto
- Tolerancia
- Honradez
- Compromiso

Página web del curso

<https://asr.github.io/courses/cm0832-set-theory/2026-1/>

Pacto Pedagógico

Página web del curso

<https://asr.github.io/courses/cm0832-set-theory/2026-1/>

Conducto regular, fechas y porcentajes de las evaluaciones

La información está en la página web del curso.

Pacto Pedagógico

Página web del curso

<https://asr.github.io/courses/cm0832-set-theory/2026-1/>

Conducto regular, fechas y porcentajes de las evaluaciones

La información está en la página web del curso.

Responsabilidad compartida

- Profesor
- Estudiantes

Asistencia a clase

Reglamento académico de los programas de posgrado, Capítulo VI, Artículo 62, Parágrafo 2.

“El estudiante de posgrado cuyas faltas de asistencia lleguen al treinta por ciento (30%) del total de las horas de clase programadas para el curso o para una parte de éste, cuando se desarrolle con más de un profesor, en secciones temáticas denominadas ‘módulo’, pierde con calificación de cero punto cero (0.0) del seminario o curso correspondiente y esta nota afecta el promedio crédito acumulado.”

Orientaciones para el curso

- Se recomienda cuatro horas de trabajo por semana (dos horas por cada hora de clase).
- Las clases son presenciales.
- La evaluación a la docencia es obligatoria.
- Se recomienda revisar periódicamente los canales de comunicación institucionales (EAFIT Interactiva y el correo institucional).
- El estudiante podrá entrar a clase a más tardar 20 minutos después de la hora asignada para su inicio.

Outline

Pácto Pedagógico

Introduction

First-Order Theories

References

Course Outline

- Formal languages and theories. Methods of proof. Naive set theory.
- Axiomatic set theory. Operations on sets.
- Relations, functions and orderings.
- Natural numbers.
- Finite, countable and uncountable sets.
- Cardinal numbers.
- Ordinal numbers.
- Axiom of choice.

Preliminaries

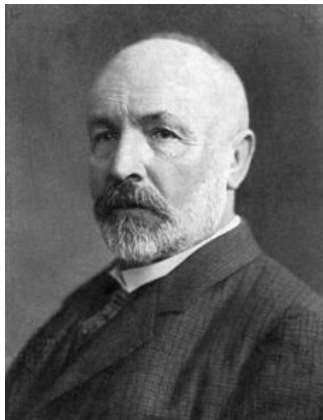
Textbook

Karel Hrbacek and Thomas Jech ([1978] 1999). Introduction to Set Theory.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Origins

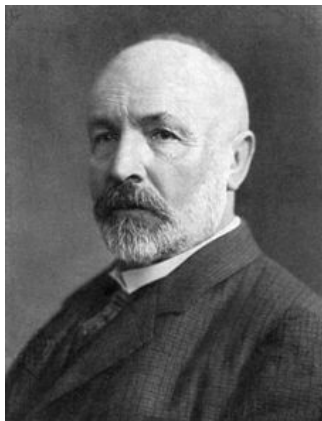


Georg Cantor (1845 – 1918)



Cantor around 1870

Origins

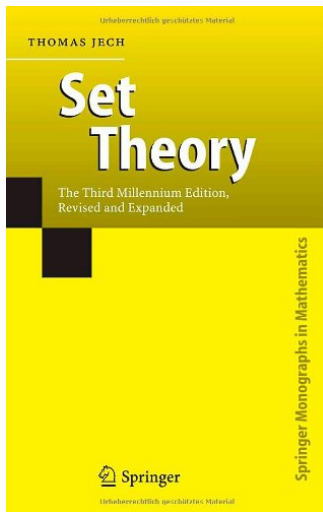


Georg Cantor (1845 – 1918)[†]

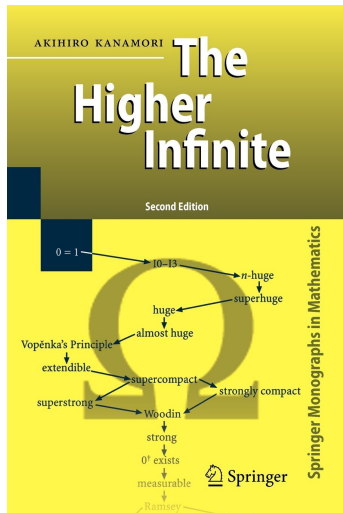


Cantor around 1870

[†]Figures source: https://en.wikipedia.org/wiki/Georg_Cantor.



*“Set theory **was invented** by Georg Cantor. . . It was however Cantor who realized the significance of one-to-one functions between sets and introduced the notion of cardinality of a set.” (Jech [1978] **2006**, p. 15)*



*"Set theory **was born** on that December 1873 day when Cantor established that the reals are uncountable, i.e. there is no one-to-one correspondence between the reals and the natural numbers." (Kanamori [1994] 2009, p. XII)*

Naive Set Theory

Remark

Cantor set theory is also called **naive** set theory.

Naive Set Theory

Cantor's set definition

*“Unter einer **Menge** verstehen wir jede Zusammenfassung M von bestimmten wohlunterschiedenen Objecten m unsrer Anschauung oder unseres Denkens (welche die **Elemente** von M genannt werden) zu einem Ganzen.”*
(Cantor 1895, p. 481)

*“By an **aggregate** (Menge) we are to understand any collection into a whole M of definite and separate objects m of our intuition or our thought. These objects are called the **elements** of M .”* (Cantor [1895] 1915, p. 85)

*“A **set** is a collection into a whole of definite, distinct objects of our intuition or our thought. The objects are called **elements (members)** of the set.”*
(Hrbacek and Jech [1978] 1999, p. 1)

Naive Set Theory

Description

*“**Sets** are not objects of the real world, like tables or stars; they are **created by our mind**, not by our hands... The human mind possesses the ability to **abstract**, to think of a variety of different objects as **being bound together** by some common **property**, and thus to form a set of objects having that property.” (p .1)*

Naive Set Theory

Description

*“**Sets** are not objects of the real world, like tables or stars; they are **created by our mind**, not by our hands... The human mind possesses the ability to **abstract**, to think of a variety of different objects as **being bound together** by some common **property**, and thus to form a set of objects having that property.” (p .1)*

Examples

In the lecture.

Naive Set Theory

Remark

A classical text in naive set theory is (Halmos 1960).

Naive Set Theory

Problem

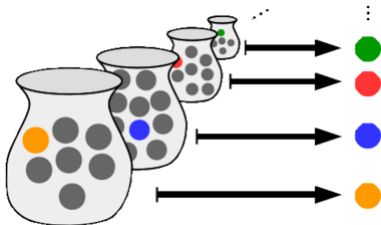
Implicit use of “theorems” on sets.

Naive Set Theory

Problem

Implicit use of “theorems” on sets.

An example of such “theorem” was the use of **axiom of choice** illustrated by the figure.

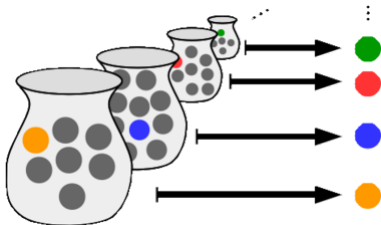


Naive Set Theory

Problem

Implicit use of “theorems” on sets.

An example of such “theorem” was the use of **axiom of choice** illustrated by the figure.



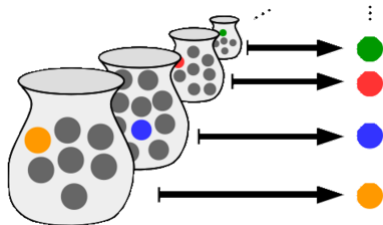
A very complete history of origins, development and influence of the axiom of choice is in (Moore 1982).

Naive Set Theory

Problem

Implicit use of “theorems” on sets.

An example of such “theorem” was the use of **axiom of choice** illustrated by the figure.[†]



A very complete history of origins, development and influence of the axiom of choice is in (Moore 1982).

[†]Figure source: <https://commons.wikimedia.org/w/index.php?curid=48219447> .

Naive Set Theory

Problem

Too general method of abstraction (i.e. axiom schema of **unrestricted** comprehension).

Naive Set Theory

Problem

Too general method of abstraction (i.e. axiom schema of **unrestricted** comprehension).

Let P be a unary property, then $\{x \mid P(x)\}$ is a set.

Naive Set Theory

Problem

Too general method of abstraction (i.e. axiom schema of **unrestricted** comprehension).

Let P be a unary property, then $\{x \mid P(x)\}$ is a set.

What is a unary property?

Naive Set Theory

Problem

Too general method of abstraction (i.e. axiom schema of **unrestricted** comprehension).

Let P be a unary property, then $\{x \mid P(x)\}$ is a set.

What is a unary property?

This problem was the cause of Russell's paradox: Let R be a “set” defined by

$$R = \{x \mid x \notin x\}.$$

The Crisis in the Foundations of Mathematics

Paradoxes $\Rightarrow$ Crisis $\Rightarrow$ $\left\{ \begin{array}{ll} \text{Logicism} & (\text{Russell and Whitehead}) \\ \text{Formalism} & (\text{Hilbert}) \\ \text{Intuitionism} & (\text{Brouwer}) \end{array} \right.$

The Crisis in the Foundations of Mathematics

Logicism (Russell and Whitehead)

“The logicistic thesis is that mathematics is a branch of logic. The mathematical notions are to be defined in terms of the logical notions. The theorems of mathematics are to be proved as theorems of logic.” (Kleene [1952] 1974, p. 43)

The Crisis in the Foundations of Mathematics

Formalism (Hilbert)

“Classical mathematics shall be formulated as a formal axiomatic theory, and this theory shall be proved to be consistent, i.e. free from contradiction.”
(Kleene [1952] 1974, p. 53)

The Crisis in the Foundations of Mathematics

Intuitionism (Brouwer)

“Intuitionism is based on the idea that mathematics is a creation of the mind. The truth of a mathematical statement can only be conceived via a mental construction that proves it to be true.” (Iemhoff 2024)

Axiomatic Set Theory

Description

*“We formulate some of the relatively simple properties of sets used by mathematicians as **axioms**, and then take care to check that all **theorems** follow logically from the axioms. Since the axioms are **obviously** true and the theorems logically follow from them, the theorems are also true (not necessarily obviously). We end up with a body of truths about sets which includes, among other things, the basic properties of natural, rational, and real numbers, functions, orderings, etc., but **as far as is known**, no contradictions.” (p. 3)*

Axiomatic Set Theory

Axiomatic set theory as a foundational system for mathematics

- *“Our axioms provide a sufficient collection of assumptions for the development of the whole of mathematics—a remarkable fact.” (Enderton 1977, p. 11)*
- *“Experience has shown that practically all notions used in contemporary mathematics can be defined, and their mathematical properties derived, in this axiomatic system. In this sense, the axiomatic set theory serves as a satisfactory foundations for the other branches of mathematics.” (p. 3)*
- *“But why axiomatize set theory in the first place? Well, for one thing, it is well known that set theory provides a unified framework for the whole of pure mathematics, and surely if anything deserves to be put on a sound basis it is such a foundational subject.” (Devlin [1979] 1993, p. 29)*
- *“Conventional mathematics is based on ZFC (the Zermelo-Fraenkel axioms, including the Axiom of Choice). Working withing ZFC, on develops: . . . All the mathematics found in basic texts on analysis, topology, algebra, etc.” (Kunen [2011] 2013, p. 1)*

Axiomatic Set Theory

Some axiomatic systems of set theory

- Zermelo-Fraenkel set theory (ZF)
- Zermelo-Fraenkel set theory with Choice (ZFC)
- von Neumann-Bernays-Gödel set theory (NBG)
- Morse-Kelley set theory (MK)
- Tarski-Grothendieck set theory (TG)

Axiomatic Set Theory

But!

“On the other hand, we do not claim that every true fact about sets can be derived from the axioms we present. The axiomatic system is not complete in this sense.” (p. 3)

Foundations of Mathematics

Foundational systems

- (a) Set theories (with additional axioms)
- (b) Category theories
- (c) Type theories
- (d) Univalent foundations
- (e) Homotopy type theories

Foundations of Mathematics

Foundational systems[†]

- (a) Set theories (with additional axioms)
- (b) Category theories
- (c) Type theories
- (d) Univalent foundations
- (e) Homotopy type theories

[†]See, e.g. (Centrone, Kant and Sarikaya 2019).

Outline

Pácto Pedagógico

Introduction

First-Order Theories

References

First-Order Theories

First-order logic: Two historical remarks

“First-order logic was explicitly identified by Peirce in 1885, but then forgotten. It was independently re-discovered in Hilbert’s 1917/18 lectures, and given wide currency in the 1928 monograph, Hilbert & Ackermann. Peirce was the first to identify it: but it was Hilbert who put the system on the map.” (Ewald 2019)

First-Order Theories

First-order logic: Two historical remarks

“First-order logic was explicitly identified by Peirce in 1885, but then forgotten. It was independently re-discovered in Hilbert’s 1917/18 lectures, and given wide currency in the 1928 monograph, Hilbert & Ackermann. Peirce was the first to identify it: but it was Hilbert who put the system on the map.” (Ewald 2019)

“Nevertheless, Hilbert did not at any point regard first-order logic as the proper basis for mathematics. . . It was in Skolem’s work on set theory (1923) that first-order logic was first proposed as all of logic and that set theory was first formulated within first-order logic.” (Moore 1988, p. 128)

First-Order Theories

Notation: Logical constants

$\wedge$	(and)	conjunction
$\vee$	(or)	(inclusive) disjunction
$\rightarrow$	(if __, then __)	conditional, material implication
$\neg$	(not)	negation
$\leftrightarrow$	(if and only if)	bi-conditional, material equivalence
$\perp$	(falsity)	bottom, falsum
$(\forall x)$	(for every x)	universal quantifier
$(\exists x)$	(there exists a x)	existential quantifier
$(\exists!x)$	(there exists one and only one x)	unique existential quantifier
$\doteq$	(equal)	identity, equality

First-Order Theories

Preliminaries logics

- First-order logic with identity
- Non-logic symbols and non-logic axioms
- Theories
- Definitions

First-Order Theories

Description

“The adjective ‘first-order’ is used to distinguish the languages we shall study here from those in which there are predicates having other predicates or functions as arguments or in which predicate quantifiers or function quantifiers are permitted, or both.” (Mendelson [1964] 2015, p. 53)

First-Order Theories

Description

“The adjective ‘first-order’ is used to distinguish the languages we shall study here from those in which there are predicates having other predicates or functions as arguments or in which predicate quantifiers or function quantifiers are permitted, or both.” (Mendelson [1964] 2015, p. 53)

Remark

For an introduction to first-order languages and first-order theories, see e.g. (Hamilton 1978) or (Mendelson [1964] 2015).

First-Order Theories

Example (first-order Dedekind-Peano arithmetic)

- Non-logical symbols

The formal language $\mathcal{L}$ of the first-order theory of arithmetic (FA) is defined by

$$\mathcal{L} = \{', +, *, 0\}, \quad \text{where}$$

- (a) the symbol $'$ is a unary function symbol (successor function),
- (b) the symbol $+$ is a binary function symbol (addition function),
- (c) the symbol $*$ is a binary function symbol (multiplication function) and
- (d) the symbol 0 is a constant symbol (zero element).

First-Order Theories

Example (first-order Dedekind-Peano arithmetic)

- Non-logical symbols

The formal language $\mathcal{L}$ of the first-order theory of arithmetic (FA) is defined by

$$\mathcal{L} = \{', +, *, 0\}, \quad \text{where}$$

- (a) the symbol $'$ is an unary function symbol (successor function),
- (b) the symbol $+$ is a binary function symbol (addition function),
- (c) the symbol $*$ is a binary function symbol (multiplication function) and
- (d) the symbol 0 is a constant symbol (zero element).

That is, the symbols $\{', +, *, 0\}$ are **primitive (undefined)** concepts in FA.

(continued on next slide)

First-Order Theories

Example (first-order Dedekind-Peano arithmetic)

- Non-logical axioms of FA

$$(\forall n)(0 \neq n') \quad (\text{FA}_1)$$

$$(\forall m)(\forall n)(m' \doteq n' \rightarrow m \doteq n) \quad (\text{FA}_2)$$

$$(\forall n)(n + 0 \doteq n) \quad (\text{FA}_3)$$

$$(\forall m)(\forall n)(m + n' \doteq (m + n)') \quad (\text{FA}_4)$$

$$(\forall n)(n * 0 \doteq 0) \quad (\text{FA}_5)$$

$$(\forall m)(\forall n)(m * n' \doteq (m * n) + m) \quad (\text{FA}_6)$$

For any unary property P ,

$$[P(0) \wedge (\forall n)(P(n) \rightarrow P(n'))] \rightarrow (\forall n)(Pn) \quad (\text{FA}_7) \text{ (axiom schema of induction)}$$

First-Order Theories

Example (first-order Dedekind-Peano arithmetic)

- Non-logical axioms of $\text{FA}^\dagger$

$$(\forall n)(0 \neq n') \quad (\text{FA}_1)$$

$$(\forall m)(\forall n)(m' \doteq n' \rightarrow m \doteq n) \quad (\text{FA}_2)$$

$$(\forall n)(n + 0 \doteq n) \quad (\text{FA}_3)$$

$$(\forall m)(\forall n)(m + n' \doteq (m + n)') \quad (\text{FA}_4)$$

$$(\forall n)(n * 0 \doteq 0) \quad (\text{FA}_5)$$

$$(\forall m)(\forall n)(m * n' \doteq (m * n) + m) \quad (\text{FA}_6)$$

For any unary property P ,

$$[P(0) \wedge (\forall n)(P(n) \rightarrow P(n'))] \rightarrow (\forall n)(Pn) \quad (\text{FA}_7) \text{ (axiom schema of induction)}$$

[†]See, e.g. (Machover 1996; Hájek and Pudlák [1993] 1998; Skolem 1955; Robinson 1949).

Outline

Pácto Pedagógico

Introduction

First-Order Theories

References

References

- Georg Cantor (1895). Beiträge zur Begründung der transfiniten Mengenlehre. *Mathematische Annalen* 46.4, pp. 481–512. DOI: [10.1007/BF02124929](https://doi.org/10.1007/BF02124929) (cit. on pp. 18, 52).
- [1895] (1915). Contributions to the Founding of the Theory of Transfinite Numbers. Translation of Cantor (1895). Translated, and provided with an introduction and notes, by Philip E. B. Jourdain. Dover Publications (cit. on p. 18).
- Stefania Centrone, Deborah Kant and Deniz Sarikaya, eds. (2019). Reflections on the Foundations of Mathematics. General Thoughts. Vol. 407. Studies in Epistemology, Logic, Methodology, and Philosophy of Science. Springer (cit. on pp. 38, 39).
- Keith Devlin [1979] (1993). The Joy of Sets. Fundamentals of Contemporary Set Theory. 2nd ed. Undergraduate Texts in Mathematics. Springer-Verlag (cit. on p. 35).
- Herbert B. Enderton (1977). Elements of Set Theory. Academic Press (cit. on p. 35).
- William Ewald (2019). The Emergence of First-Order Logic. In: The Stanford Encyclopedia of Philosophy. Ed. by Edward N. Zalta. Spring 2019. Metaphysics Research Lab, Stanford University. URL: <https://plato.stanford.edu/archives/spr2019/entries/logic-firstorder-emergence/> (visited on 18/03/2023) (cit. on pp. 41, 42).

References

- Petr Hájek and Pavel Pudlák [1993] (1998). Metamathematics of First-Order Arithmetic. Second printing. Springer (cit. on pp. 49, 50).
- Paul R. Halmos (1960). Naive Set Theory. The University Series in Undergraduate Mathematics. Van Nostrand Reinhold Company (cit. on p. 21).
- A. G. Hamilton (1978). Logic for Mathematicians. Cambridge University Press (cit. on pp. 45, 46).
- Karel Hrbacek and Thomas Jech [1978] (1999). Introduction to Set Theory. Third Edition, Revised and Expanded. Marcel Dekker (cit. on pp. 12, 18).
- Rosalie Iemhoff (2024). Intuitionism in the Philosophy of Mathematics. In: The Stanford Encyclopedia of Philosophy. Ed. by Edward N. Zalta and Uri Nodelman. Summer 2024. Metaphysics Research Lab, Stanford University. URL: <https://plato.stanford.edu/archives/sum2024/entries/intuitionism> (cit. on p. 33).
- Thomas J. Jech [1978] (2006). Set Theory. The Third Millennium Edition, Revised and Expanded. Corrected 4th printing. Springer (cit. on p. 15).
- Akihiro Kanamori [1994] (2009). The Higher Infinite. Large Cardinals in Set Theory from Their Beginnings. 2nd ed. Springer (cit. on p. 16).

References

- Stephen Cole Kleene [1952] (1974). Introduction to Metamathematics. Seventh reprint. Bibliotheca Mathematica: A Series of Monographs on Pure and Applied Mathematics. Volume 1. North-Holland (cit. on pp. 31, 32).
- Kenneth Kunen [2011] (2013). Set Theory. Revised edition. Vol. 34. Mathematical Logic and Foundations. College Publications (cit. on p. 35).
- Moshé Machover (1996). Set Theory, Logic and their Limitations. Cambridge University Press (cit. on pp. 49, 50).
- Elliott Mendelson [1964] (2015). Introduction to Mathematical Logic. 6th ed. CRC Press (cit. on pp. 45, 46).
- Gregory H. Moore (1982). Zermelo's Axiom of Choice. Its Origins, Development, and Influence. Vol. 8. Studies in the History of Mathematics and Physical Sciences. Springer-Verlag. DOI: 10.1007/978-1-46 (cit. on pp. 22–25).
- (1988). The Emergence of First-Order Logic. In: History and Philosophy of Modern Mathematics. Ed. by William Aspray and Philip Kitcher. Vol. 11. Minnesota Studies in the Philosophy of Science. University of Minnesota Press, pp. 95–134. URL: <https://hdl.handle.net/11299/185662> (cit. on pp. 41, 42).

References

- Julia Robinson (1949). Definability and Decision Problems in Arithmetic. *The Journal of Symbolic Logic* 14.2, pp. 98–114. DOI: [10.2307/2266510](https://doi.org/10.2307/2266510) (cit. on pp. 49, 50).
- Th. Skolem (1955). Peano's Axioms and Models of Arithmetic. In: *Mathematical Interpretation of Formal Systems*. Ed. by Th. Skolem et al. Vol. 16. *Studies in Logic and the Foundations of Mathematics*. North-Holland Publishing Company, pp. 1–14 (cit. on pp. 49, 50).